THE FIVE SMALL QUASI-COHERENT DERIVED CATEGORIES OF A QUASI-COMPACT QUASI-SEPARATED SCHEME AGREE

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Allowed standard prerequisites (editorial).

- The category of sheaves on the small Zariski site of quasi-coherent modules agrees with the quasi-coherent modules on the ringed space, and the categories of modules agree, so the two derived categories are equivalent. invoked at proof lines 11019-11025. Gives the equivalence between the first two categories of the statement.
- The equivalence of ringed topoi between the site of affines and the Zariski site induces an equivalence of module categories that preserves the quasi-coherent subcategories because intrinsicness of that notion. invoked at proof lines 11026-11032. Gives the equivalence between the second and third categories.
- The comparison lemma for adequate modules produces an equivalence of derived categories from a morphism of ringed sites once four conditions on a basis and a subcategory of presheaves are checked. invoked at proof lines 11033-11060. Supplies the equivalence between the last two categories once its hypotheses are verified.
- Quasi-coherent modules on the affine site are exactly the presheaves compatible with localizing the ring of sections, and they are also exactly the quasi-coherent modules in the chaotic topology. used at proof lines 11045-11058. Identifies the two descriptions of the subcategory to which the comparison lemma is applied.

Source reference. Original lemma 0GZR: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 11015-11017 for the equivalence between the last two categories; the flatness of the ring maps that makes it applicable is verified inside the counted span.

Source reference. Original remark 03FG: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 11019-11021 for the identification of the module categories; the subsequent correspondence of quasi-coherent subcategories is inside the counted span.

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Source reference. Original proposition 03DX: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 11022-11024 to conclude the equivalence of the first two categories; everything else used for it is inside the counted span.

Source reference. Original lemma 03DM: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 11029-11031 to see that the equivalence of module categories preserves quasi-coherent subcategories; the conclusion drawn is inside the counted span.

Source reference. Original lemma 07A8: sources/stacks-chow/tags.txt, source lines 1–1. Applied at proof lines 11033-11060 to produce the equivalence between the last two categories; the choice of basis and subcategory and the verification of the four conditions are inside the counted span.

Source reference. Original lemma 0GZX: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 11046-11048 to identify the members of the chosen subcategory with the quasi-coherent sheaves on the Zariski topology; the identification with the chaotic topology case follows inside the counted span.

Source reference. Original lemma 0GZN: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 11049-11053 for the second description of the subcategory; the conclusion that both descriptions agree is inside the counted span.

Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 11084-11086 for the fourth condition of the comparison lemma; the other three conditions are verified inside the counted span.

Source reference. Original section 01LA: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 11072-11076 for the second condition of the comparison lemma; the reduction to that discussion is inside the counted span.

Lemma 0.2. *In the situation above there are canonical exact equivalences between the following triangulated categories*

(1) $D_{QCoh}(\mathcal{O}_X)$,
(2) $D_{QCoh}(X_{Zar}, \mathcal{O})$,
(3) $D_{QCoh}(X_{affine,Zar}, \mathcal{O})$,
(4) $D_{QCoh}(X_{affine}, \mathcal{O}_X)$, *and*
(5) $QC(X_{affine}, \mathcal{O})$.

Proof. If $U \subset V \subset X$ are affine open, then the ring map $\mathcal{O}(V) \to \mathcal{O}(U)$ is flat. Hence the equivalence between (4) and (5) is a special case of Cohomology on Sites, Lemma 0GZR (the proof also clarifies the statement).

The ringed site $(X_{Zar}, \mathcal{O})$ and the ringed space $(X, \mathcal{O}_X)$ have the same categories of modules by Descent, Remark 03FG. Via this equivalence the quasi-coherent modules correspond by Descent, Proposition 03DX. Hence we get a canonical exact equivalence between the triangulated categories in (1) and (2).

The discussion preceding the lemma shows that we have an equivalence of ringed topoi $(Sh(X_{affine,Zar}), \mathcal{O}) \to (Sh(X_{Zar}), \mathcal{O})$ and hence an equivalence between abelian categories of modules. Since the notion of quasi-coherent modules is intrinsic (Modules on Sites, Lemma 03DM) we see that this equivalence preserves the subcategories of quasi-coherent modules. Thus we get a canonical exact equivalence between the triangulated categories in (2) and (3).

To get an exact equivalence between the triangulated categories in (3) and (4) we will apply Cohomology on Sites, Lemma 07A8 to the morphism $\epsilon : (X_{affine,Zar}, \mathcal{O}) \to (X_{affine}, \mathcal{O})$ above. We take $\mathcal{B} = \mathrm{Ob}(X_{affine})$ and we take $\mathcal{A} \subset PMod(X_{affine}, \mathcal{O})$ to be the full subcategory of those presheaves $\mathcal{F}$ such that $\mathcal{F}(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U) \to \mathcal{F}(U)$ is an isomorphism. Observe that by Descent, Lemma 0GZX objects of $\mathcal{A}$ are exactly those sheaves in the Zariski topology which are quasi-coherent modules on $(X_{affine,Zar}, \mathcal{O})$. On the other hand, by Modules on Sites, Lemma 0GZN, the objects of $\mathcal{A}$ are exactly the quasi-coherent modules on $(X_{affine}, \mathcal{O})$, i.e., in the chaotic topology. Thus if we show that Cohomology on Sites, Lemma 07A8 applies, then we do indeed get the canonical equivalence between the categories of (3) and (4) using ϵ^* and $R\epsilon_*$.

We have to verify 4 conditions:

(1) Every object of $\mathcal{A}$ is a sheaf for the Zariski topology. This we have seen above.
(2) $\mathcal{A}$ is a weak Serre subcategory of $Mod(X_{affine,Zar}, \mathcal{O})$. Above we have seen that $\mathcal{A} = QCoh(X_{affine,Zar}, \mathcal{O})$ and we have seen above that these, via the equivalence $Mod(X_{affine,Zar}, \mathcal{O}) = Mod(X, \mathcal{O}_X)$, correspond to the quasi-coherent modules on X. Thus the result by the discussion in Schemes, Section 01LA.
(3) Every object of X_{affine} has a covering in the chaotic topology whose members are elements of $\mathcal{B}$. This holds because $\mathcal{B}$ contains all objects.
(4) For every object U of X_{affine} and $\mathcal{F}$ in $\mathcal{A}$ we have $H^p_{Zar}(U, \mathcal{F}) = 0$ for $p > 0$. This holds by the vanishing of cohomology of quasi-coherent modules on affines, see Cohomology of Schemes, Lemma 01XB.

This finishes the proof. □